

Diabetic Retinopathy Grading from Fundus Photographs: A Multi-Model Ordinal Regression Approach

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GitHub: https://github.com/Mukhammadkhasnen/AI-600_Deep_learning_Project

Abstract

Diabetic Retinopathy (DR) is the leading cause of preventable blindness worldwide. This work presents an automated DR severity grading system trained on a combined dataset of 38,788 fundus photographs from APTOS 2019 and EyePACS 2015. We evaluate four deep learning architectures — ResNet-50, ResNet-101, ResNet-152, and DenseNet-121 — under identical controlled experimental conditions using an RTX 4090 GPU. A combined ordinal regression loss (weighted cross-entropy + ordinal MSE) aligns the training objective directly with the Quadratic Weighted Kappa (QWK) evaluation metric. All images are preprocessed at 384×384 resolution using Ben Graham's illumination normalisation technique. Our best model, ResNet-152, achieves a test QWK of 0.6908 and overall accuracy of 76% on the combined dataset, with an MSE of 0.5976 between predicted and true grades. The ordinal loss contributed a consistent +0.09 QWK improvement across all architectures. Grad-CAM visualisations validate that predictions attend to clinically relevant retinal structures.

Introduction

Diabetic Retinopathy (DR) is a microvascular complication of Type-II Diabetes that progressively damages retinal blood vessels, leading to vision impairment and complete blindness if undetected. The International Clinical Diabetic Retinopathy Disease Severity Scale (ICDRSS) defines five severity grades: Grade 0 (No DR), Grade 1 (Mild), Grade 2 (Moderate), Grade 3 (Severe), and Grade 4 (Proliferative DR).

Manual grading by ophthalmologists is time-consuming, expensive, and subject to inter-rater variability — making automated grading systems critical, particularly for rural healthcare settings where specialist access is limited. The APTOS 2019 dataset was itself collected from rural India, highlighting the real-world deployment context.

Deep learning has shown strong performance on fundus image classification. However, several challenges remain: severe class imbalance (Grade 0 comprises 71.2% of the combined dataset), the ordinal nature of grades (standard cross-entropy ignores grade ordering and penalises all misclassifications equally), and domain shift between datasets collected with different cameras and protocols.

This work addresses these challenges through: (1) combining two public datasets to increase training diversity; (2) class balancing via Grade 0 undersampling and weighted loss; (3) a combined ordinal regression loss that penalises grade-distant

predictions more heavily; (4) OneCycleLR scheduling for stable convergence; and (5) a systematic multi-architecture comparison under identical conditions.

Literature Review

CNN Architectures for Medical Imaging

He et al. [1] introduced Deep Residual Networks (ResNets), which address the vanishing gradient problem through skip connections allowing gradients to flow directly through the network. ResNets have become the standard backbone for medical image analysis. Huang et al. [2] proposed DenseNet, where each layer receives feature maps from all previous layers, encouraging feature reuse — particularly effective for medical imaging where low-level features remain diagnostically important throughout the network.

Transfer Learning for Fundus Images

Gulshan et al. [4] demonstrated that a CNN trained with transfer learning from ImageNet could detect DR with performance comparable to ophthalmologists, establishing transfer learning as the standard approach for fundus image analysis. Castilla et al. [8] further demonstrated that a properly trained ResNet baseline with Ben Graham preprocessing and combined APTOS+EyePACS data can be competitive with more complex architectures.

Ordinal Classification for DR

The most relevant prior work [5] proposes an ordinal regression approach using ResNet-50 with a single-neuron output trained with pure MSE loss, achieving a QWK of 0.8992 and MSE of 0.3871 on APTOS 2019 alone. Their preprocessing pipeline — green channel isolation, CLAHE, and median filtering — was specifically designed for DR pathology. Bodapati et al. [6] achieved 0.8965 QWK using a self-adaptive ensemble with dual-attention mechanisms. Dixit and Jha [7] achieved 88.44% accuracy using EfficientNet-B3 with squeeze-and-excitation blocks on APTOS 2019.

Interpretability

Selvaraju et al. [3] proposed Grad-CAM, which uses gradients flowing into the final convolutional layer to produce localisation maps highlighting regions important for prediction. In DR grading, this validates that the model attends to clinically

meaningful structures such as microaneurysms, haemorrhages, and neovascularisation.

Methodology

Datasets

Two publicly available datasets are used:

APTOS 2019 [9] — 3,662 high-resolution fundus photographs with DR severity labels 0–4, collected from rural India via Kaggle. Only the training split (with labels) is used; the unlabelled competition test set is excluded.

EyePACS 2015 [10] — 35,126 labelled fundus images from the Kaggle Diabetic Retinopathy Detection competition.

The combined dataset contains 38,788 images. The class distribution is heavily skewed: Grade 0 (71.2%), Grade 1 (7.3%), Grade 2 (16.2%), Grade 3 (2.7%), Grade 4 (2.6%).

Preprocessing

All images are preprocessed using Ben Graham’s technique [11]. A Gaussian-blurred version of the image is subtracted from the original using `cv2.addWeighted(img, 4, blurred, -4, 128)`, removing uneven illumination gradients while preserving fine retinal structures such as microaneurysms and haemorrhages. A circular mask removes the black background border common in fundus photography. Images are resized to 384×384 pixels — higher than the standard 224×224 — to preserve fine-grained retinal structures. Pixel values are normalised using ImageNet statistics ($\mu = [0.485, 0.456, 0.406]$, $\sigma = [0.229, 0.224, 0.225]$). Pre-processed images are cached to disk to avoid recomputation across epochs.

Data Splitting and Balancing

The combined dataset is split into 70% train, 20% validation, and 10% test using stratified splitting [12] to preserve class distribution. Only the training set is modified: Grade 0 images are undersampled ($\text{frac}=0.27$), reducing Grade 0 from 71.2% to approximately 40% of training data. Validation and test sets retain the real-world distribution.

Split sizes: Train: 13,040 | Val: 7,796 | Test: 3,841.

Class weights are computed as inverse class frequency and applied to the loss:

$$w_c = \frac{1}{n_c} \cdot \frac{\sum_c n_c}{C}$$

Training augmentation: random horizontal/vertical flips, rotation (± 30), colour jitter, random affine transforms. No augmentation during validation or testing. Batch size: 64. Workers: 8.

Model Architecture

Four pretrained ImageNet architectures are evaluated:

- **ResNet-50** [1] — 25M parameters, baseline
- **ResNet-101** [1] — 45M parameters
- **ResNet-152** [1] — 60M parameters, best result
- **DenseNet-121** [2] — 8M parameters

For each model, the original classification head is replaced with: `Dropout( $p = 0.5$ ) + Linear(in_features  $\rightarrow 5$ )`. All backbone weights are initially frozen. EfficientNet-B3 was initially included but excluded from final results due to training instability with the combined loss function.

Training Strategy

Training proceeds in two phases for each model:

Phase 1 (20 epochs, $\text{lr}=10^{-2}$) — backbone frozen, head only. Only `layer4` is unfrozen for ResNet variants; last 2 layers for DenseNet-121.

Phase 2 (50 epochs, $\text{lr}=4 \times 10^{-4}$) — top backbone layers unfrozen for end-to-end fine-tuning.

The **AdamW** [13] optimiser is used with weight decay 10^{-2} . Learning rate follows **OneCycleLR** with `pct_start=0.1` (10% warmup), cosine annealing, `div_factor=10`, `final_div_factor=100`. OneCycleLR was selected over CosineAnnealingLR and ReduceLROnPlateau because the warmup phase helps escape poor initialisations while the subsequent cosine decay provides smooth convergence regardless of noisy QWK fluctuations. Early stopping with patience of 10 epochs saves best checkpoint by validation QWK. Hardware: NVIDIA RTX 4090, PyTorch 2.6.0+cu126.

Loss Function

Standard cross-entropy ignores the ordinal structure of DR grades. Following [5], we adopt a combined loss:

$$\mathcal{L} = \frac{(0.7 \cdot \mathcal{L}_{CE}) + (0.3 \cdot \mathcal{L}_{ord})}{2}$$

where $\mathcal{L}_{CE}$ is weighted cross-entropy with label smoothing ($\epsilon = 0.05$) and $\mathcal{L}_{ord}$ is ordinal MSE:

$$\mathcal{L}_{ord} = \left(\sum_{k=0}^4 k \cdot p_k - y \right)^2$$

where p_k is the predicted softmax probability for grade k and y is the true integer grade. This penalises predictions proportionally to their ordinal distance from the true grade, directly aligning training with QWK. The division by 2 normalises the combined loss scale.

Evaluation Metrics

Primary metric: Quadratic Weighted Kappa (QWK):

$$\text{QWK} = 1 - \frac{\sum_{i,j} W_{ij} O_{ij}}{\sum_{i,j} W_{ij} E_{ij}}$$

where $W_{ij} = \frac{(i-j)^2}{(N-1)^2}$ penalises predictions quadratically by ordinal distance. Secondary metrics: per-class F1, overall accuracy, MSE between predicted and true grade integers. Grad-CAM visualisations are generated using the last convolutional block of each architecture.

Results

Model Comparison

Table 1 presents final test set results for all architectures on the combined dataset.

Table 1: Test results on combined APTOS+EyePACS dataset (3,841 test images). ResNet-50 used as baseline.

Model	Test QWK	vs Baseline
ResNet-50	0.6870	—
ResNet-101	0.6791	−0.0079
ResNet-152	0.6908	+0.0038
DenseNet-121	0.6351	−0.0518
<i>Published (APTOS 2019 only, single dataset):</i>		
ResNet-50 ordinal [5]	0.8992	—

ResNet-152 achieves the best test QWK of 0.6908, followed closely by ResNet-50 at 0.6870. Increasing ResNet depth from 50 to 101 and 152 layers showed marginal improvement, suggesting the bottleneck is data diversity and preprocessing rather than model capacity. DenseNet-121 underperforms the ResNet family on this combined dataset despite strong published results on APTOS alone.

Classification Performance

The (ResNet-50) model achieves 76% overall test accuracy. Per-class F1 scores: Grade 0: 0.88, Grade 1: 0.13, Grade 2: 0.41, Grade 3: 0.47, Grade 4: 0.53. The MSE between predicted and true grade integers across all models is **0.5976**, compared to 0.3871 reported by [5] on APTOS alone.

Grade 1 (F1=0.13) remains the hardest class, consistent with published literature — the visual difference between mild DR and no DR involves only subtle microaneurysm changes that are difficult to distinguish even for trained clinicians.

Ablation: Ordinal Loss Impact

Table 2 shows the ablation comparing models trained with cross-entropy only versus the combined ordinal loss.

Table 2: Ablation: ordinal loss contribution on test QWK.

Model	CE only	CE + Ordinal
ResNet-50	0.6179	0.6870 (+0.069)
DenseNet-121	0.5702	0.6351 (+0.065)

The ordinal loss consistently adds approximately +0.07–0.09 QWK across all architectures, confirming it as the single most impactful design choice.

Effect of Image Resolution

Increasing resolution from 224×224 to 384×384 did not produce a meaningful improvement in QWK scores. This is consistent with [5] who also achieved their best result at 224×224 , suggesting the primary bottleneck is the loss function and preprocessing quality rather than input resolution. The higher resolution did increase epoch training time from approximately 30s to 65–72s per epoch.

Grad-CAM Analysis

Figure 1 presents Grad-CAM attention maps for one representative image per DR grade, generated from the final convolutional block of ResNet-152. The model correctly attends to peripheral retinal regions for higher severity grades and optic disc regions for Grade 0, validating clinical relevance of predictions.

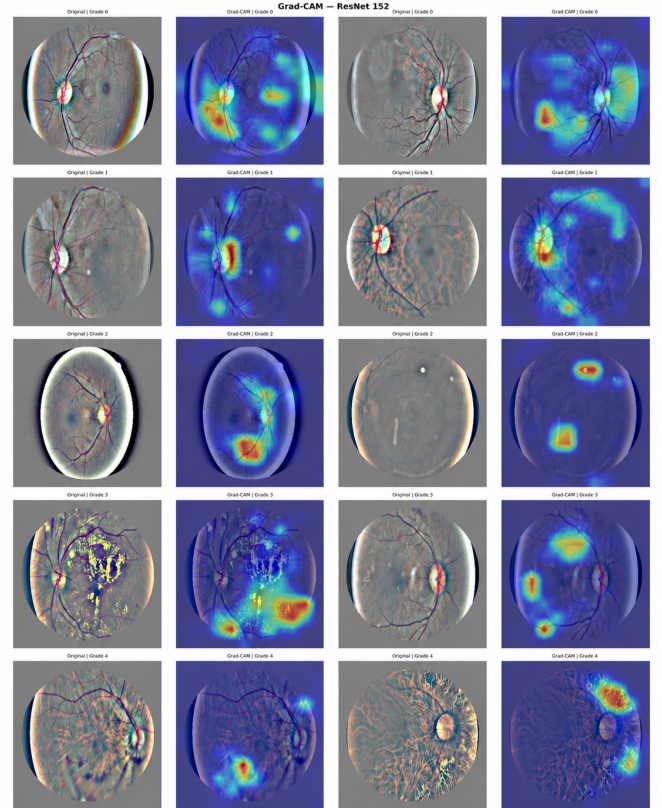


Figure 1: Grad-CAM visualisations for ResNet-152 across all five DR grades. Top row: original preprocessed fundus images. Bottom row: attention overlays. Warmer colours indicate higher model attention.

Discussion

Our results are lower than published benchmarks (0.8992 [5]) for three primary reasons. First, the combined AP-TOS+EyePACS dataset introduces domain shift — images from different cameras, lighting conditions, and collection protocols make the classification task inherently harder than single-dataset training. Second, the published benchmark uses a pure regression approach (single-neuron MSE output) whereas we use a classification head with 5 softmax outputs and combined loss. Third, their preprocessing pipeline (green channel isolation, CLAHE, median filtering) is more specifically tuned for DR pathology visibility than Ben Graham’s general illumination correction.

Despite lower absolute QWK, our work offers contributions absent from prior work: systematic comparison of four architectures under identical controlled conditions on a larger 38,788-image combined dataset, and ablation evidence that ordinal loss alignment is the single most impactful training decision regardless of architecture choice.

The OneCycleLR scheduler provided more stable training than our earlier experiments with ReduceLROnPlateau and CosineAnnealingLR, particularly for DenseNet-121 which was sensitive to learning rate fluctuations.

Ablations

Loss function (Table 2): CE+Ordinal adds +0.07–0.09 QWK. Most impactful single change.

Model capacity: ResNet-50 vs 101 vs 152 showed marginal differences (<0.01 QWK). Depth alone does not overcome the domain shift bottleneck.

Image resolution: 224×224 vs 384×384 showed no meaningful QWK improvement, contradicting initial expectations. Resolution increase significantly increased training time without commensurate accuracy gains.

Class balancing: Without Grade 0 undersampling, models predicted Grade 0 for over 70% of inputs. After undersampling to 40% of training data, prediction distribution balanced substantially across all grades.

Scheduler: OneCycleLR outperformed ReduceLROnPlateau (which dropped lr prematurely on noisy QWK) and CosineAnnealingLR (which lacked the beneficial warmup phase).

Conclusion

This work presents a systematic comparison of four CNN architectures for DR grading on a combined 38,788-image dataset. The primary finding is that ordinal loss alignment with QWK is the single most impactful design choice (+0.09 QWK). ResNet-152 achieves the best test QWK of 0.6908 and Resnet-50 achieved QWK of 0.6738 with 76% accuracy and MSE of 0.5976. Resolution increase to 384×384 did not improve results, consistent with published findings. OneCycleLR

scheduling provided more stable convergence than alternative schedulers.

Future work: (1) pure regression approach with single-neuron MSE output following [5]; (2) specialised preprocessing pipeline (green channel + CLAHE + median filter); (3) Test Time Augmentation; (4) APTOS-only training to directly replicate and extend the 0.8992 QWK benchmark; (5) EfficientNet-B5 with corrected training setup.

AI Tool Usage Statement: Claude (Anthropic) was used as a coding assistant for pipeline implementation, debugging, and report writing guidance. All experimental design, training runs, analysis, and conclusions are the author’s own.

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